Lecture 7:

Module 1: Cloud Concepts Overview

Module 2: Cloud Economics and Billing

Module 1: Cloud Concepts Overview

AWS Academy Cloud Foundations

Module overview

Topics

- Introduction to cloud computing
- Advantages of cloud computing
- Introduction to Amazon Web Services (AWS)
- AWS Cloud Adoption Framework (AWS CAF)

Knowledge check

Module objectives

After completing this module, you should be able to:

- Define different types of cloud computing models
- Describe six advantages of cloud computing
- Recognize the main AWS service categories and core services
- Review the AWS Cloud Adoption Framework (AWS CAF)

Section 1: Introduction to cloud computing

Module 1: Cloud Concepts Overview

What is cloud computing?

Cloud computing defined

Cloud computing is the **on-demand** delivery of compute power, database, storage, applications, and other IT resources **via the internet** with **pay-as-you-go** pricing.

Infrastructure as software

Cloud computing enables you to **stop thinking of your infrastructure as hardware**, and instead **think of (and use) it as software.**

Traditional computing model

- Infrastructure as hardware
- Hardware solutions:
  - Require space, staff, physical security, planning, capital expenditure
  - Have a long hardware procurement cycle
  - Require you to provision capacity by guessing theoretical maximum peaks

Cloud computing model

- Infrastructure as software
- Software solutions:
  - Are flexible
  - Can change more quickly, easily, and cost-effectively than hardware solutions
  - Eliminate the undifferentiated heavy-lifting tasks

Cloud service models

IaaS
(infrastructure as a service)

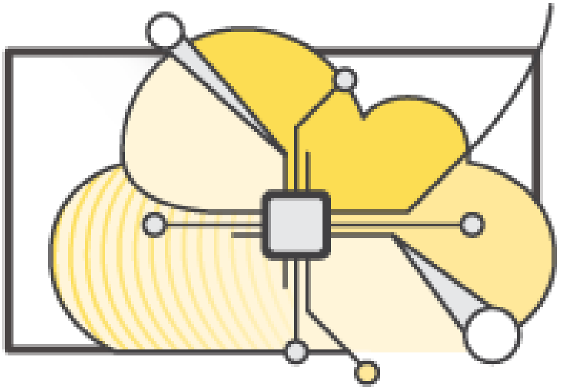
PaaS
(platform as a service)

SaaS
(software as a service)

More control
over IT resources

Less control
over IT resources

Cloud computing deployment models

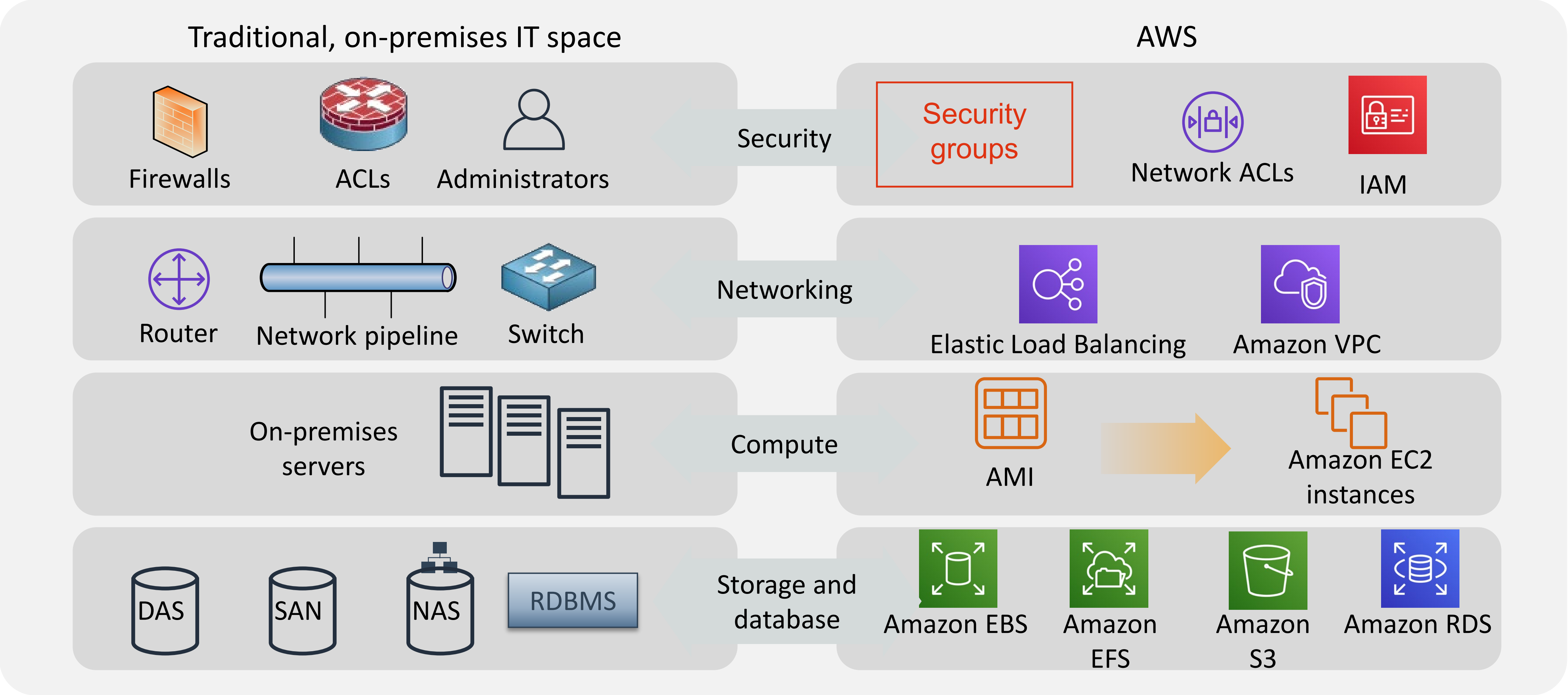


Cloud

Hybrid

On-premises
(private cloud)

Similarities between AWS and traditional IT



Section 1 key takeaways

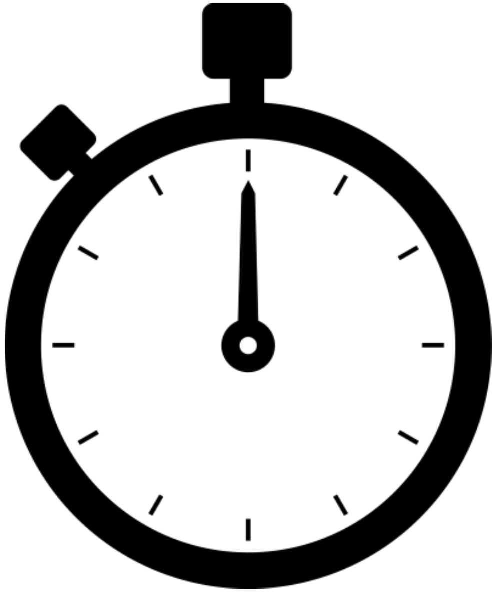
- Cloud computing is the on-demand delivery of IT resources via the internet with pay-as-you-go pricing.
- Cloud computing enables you to think of (and use) your infrastructure as software.
- There are three cloud service models: IaaS, PaaS, and SaaS.
- There are three cloud deployment models: cloud, hybrid, and on-premises or private cloud.
- Almost anything you can implement with traditional IT can also be implemented as an AWS cloud computing service.

Section 2: Advantages of cloud computing

Module 1: Cloud Concepts Overview

Trade capital expense for variable expense

Data center investment based on forecast



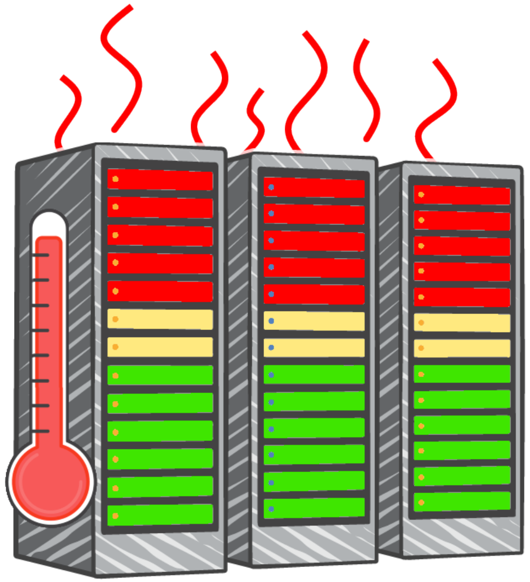
Pay only for the amount you consume

Massive economies of scale

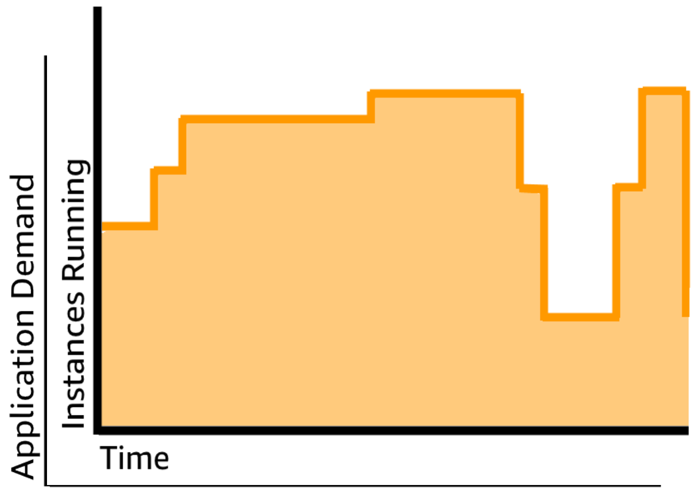
Because of aggregate usage from all customers, AWS can achieve higher economies of scale and pass savings on to customers.

Stop guessing capacity

Overestimated server capacity



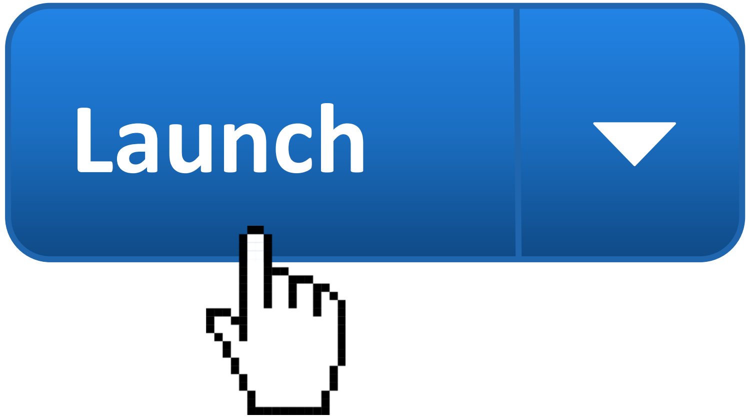
Underestimated server capacity



Scaling on demand

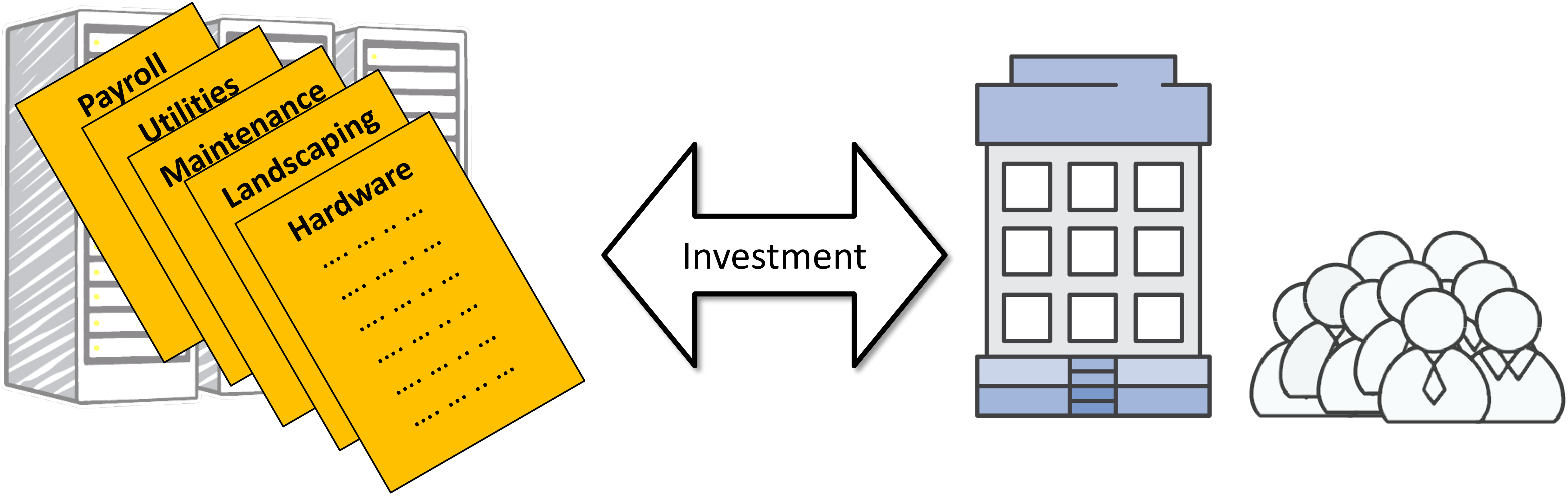
Increase speed and agility

Weeks between wanting resources and having resources



Minutes between wanting resources and having resources

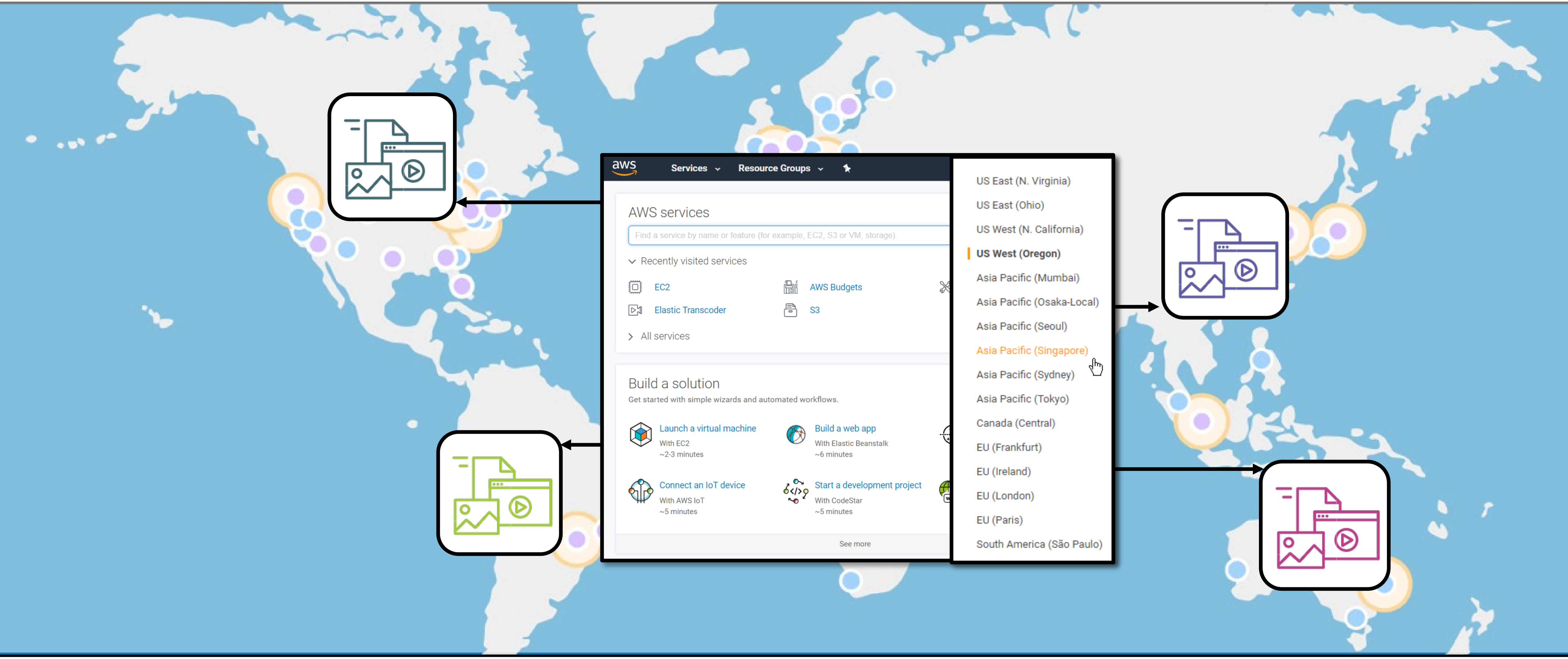
Stop spending money on running and maintaining data centers



Running data centers

Business and customers

Go global in minutes



Section 2 key takeaways

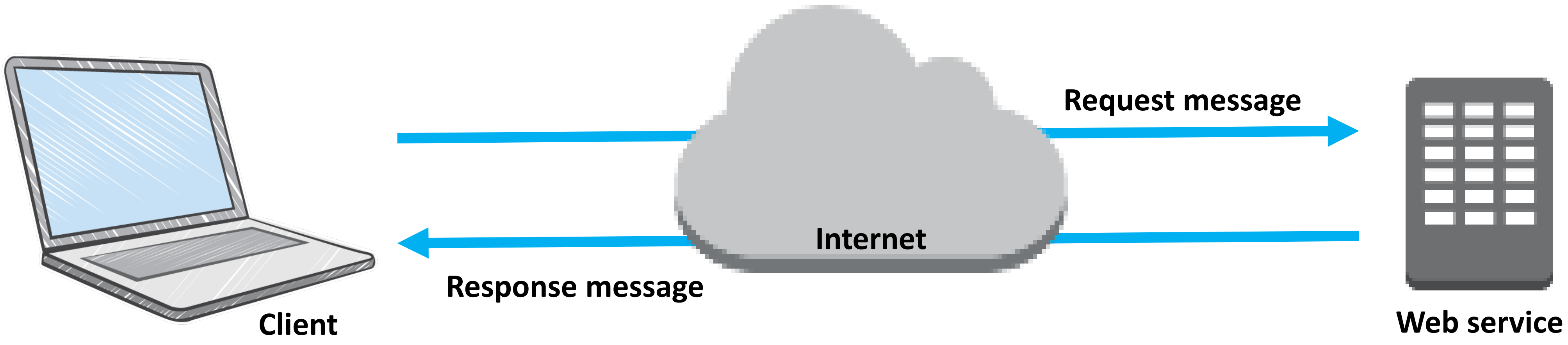
- Trade capital expense for variable expense
- Benefit from massive economies of scale
- Stop guessing capacity
- Increase speed and agility
- Stop spending money on running and maintaining data centers
- Go global in minutes

Section 3: Introduction to Amazon Web Services (AWS)

Module 1: Cloud Concepts Overview

What are web services?

A **web service** is any piece of software that makes itself available over the internet and uses a **standardized format**—such as Extensible Markup Language (XML) or JavaScript Object Notation (JSON)—for the request and the response of an **application programming interface (API) interaction**.



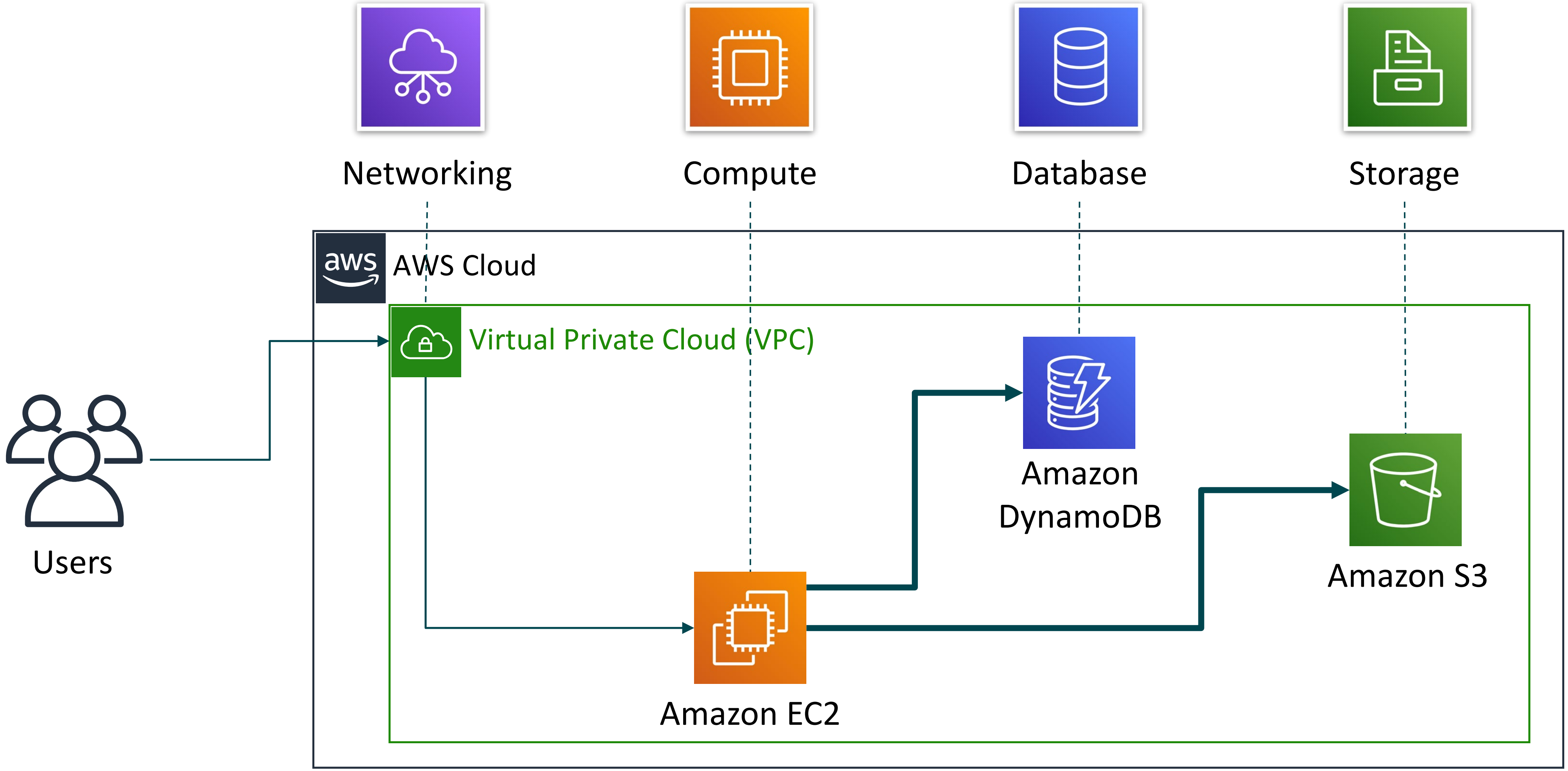
What is AWS?

- AWS is a **secure cloud platform** that offers a **broad set of global cloud-based products**.
- AWS provides you with **on-demand access** to compute, storage, network, database, and other IT resources and management tools.
- AWS offers **flexibility**.
- You **pay only for the individual services you need**, for **as long as you use them**.
- AWS services **work together** like building blocks.

Categories of AWS services

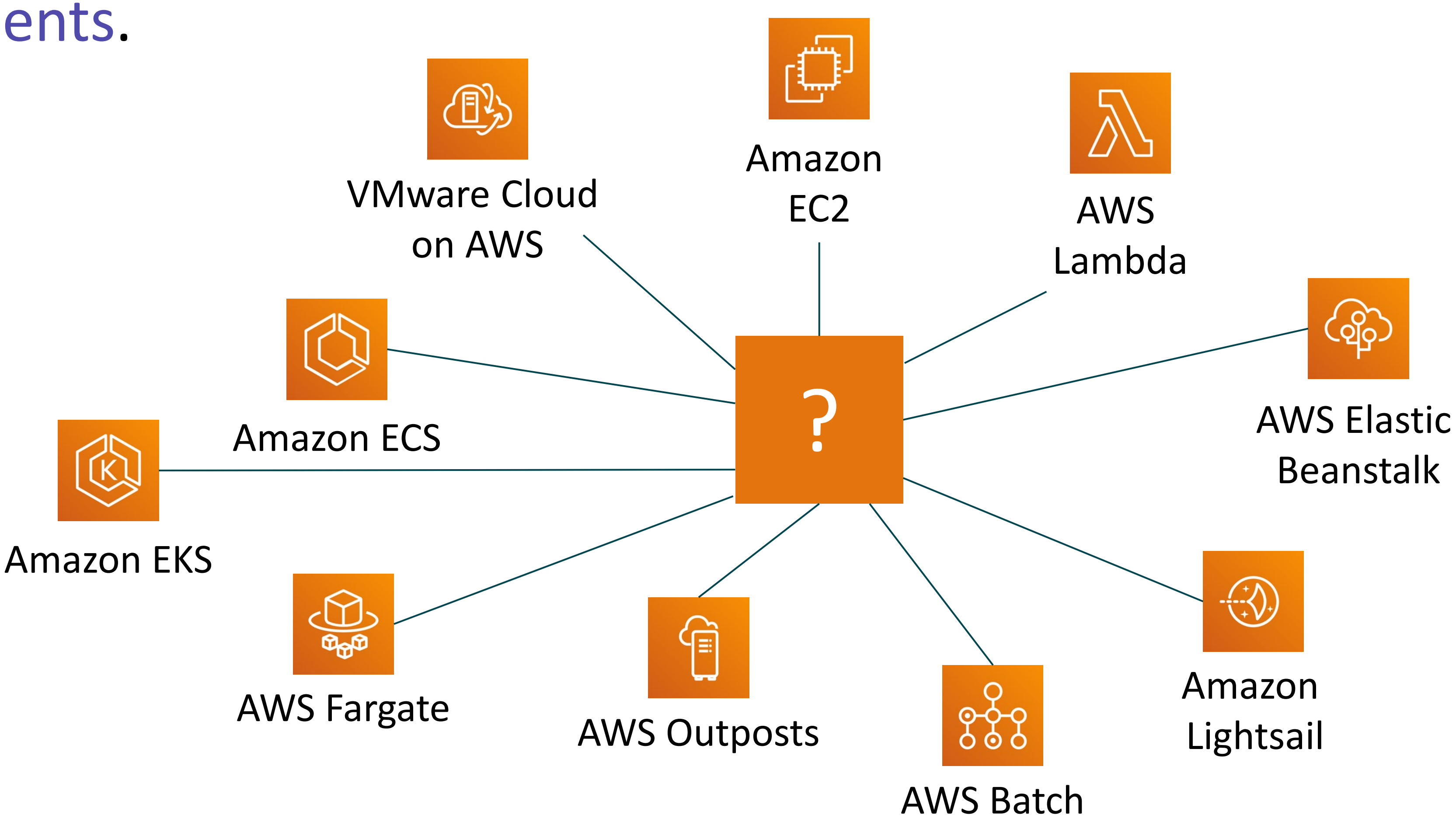
- Analytics
- Application Integration
- AR and VR
- Blockchain
- Business Applications
- Compute
- Cost Management
- Customer Engagement
- Database
- Developer Tools
- End User Computing
- Game Tech
- Internet of Things
- Machine Learning
- Management and Governance
- Media Services
- Migration and Transfer
- Mobile
- Networking and Content Delivery
- Robotics
- Satellite
- Security, Identity, and Compliance
- Storage

Simple solution example



Choosing a service

The service you select depends on your business goals and technology requirements.



Services covered in this course

Compute services –

- Amazon EC2
- AWS Lambda
- AWS Elastic Beanstalk
- Amazon EC2 Auto Scaling
- Amazon ECS
- Amazon EKS
- Amazon ECR
- AWS Fargate

Security, Identity, and Compliance services –

- AWS IAM
- Amazon Cognito
- AWS Shield
- AWS Artifact
- AWS KMS

Storage services –

- Amazon S3
- Amazon S3 Glacier
- Amazon EFS
- Amazon EBS

Database services –

- Amazon RDS
- Amazon DynamoDB
- Amazon Redshift
- Amazon Aurora

Networking and Content Delivery services –

- Amazon VPC
- Amazon Route 53
- Amazon CloudFront
- Elastic Load Balancing

Management and Governance services –

- AWS Trusted Advisor
- AWS CloudWatch
- AWS CloudTrail
- AWS Well-Architected Tool
- AWS Auto Scaling
- AWS Command Line Interface
- AWS Config
- AWS Management Console
- AWS Organizations

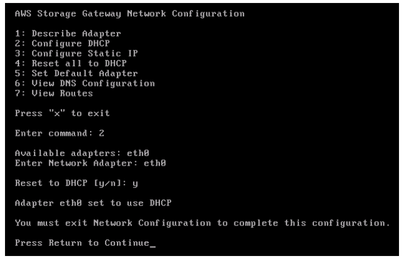
AWS Cost Management services –

- AWS Cost & Usage Report
- AWS Budgets
- AWS Cost Explorer

Three ways to interact with AWS

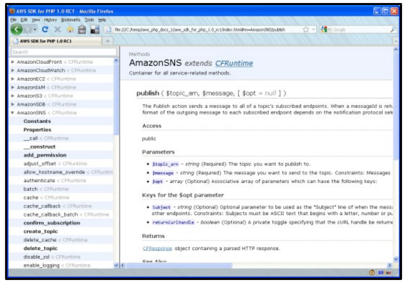
AWS Management Console

Easy-to-use graphical interface



Command Line Interface (AWS CLI)

Access to services by discrete commands or scripts



Software Development Kits (SDKs)

Access services directly from your code (such as Java, Python, and others)

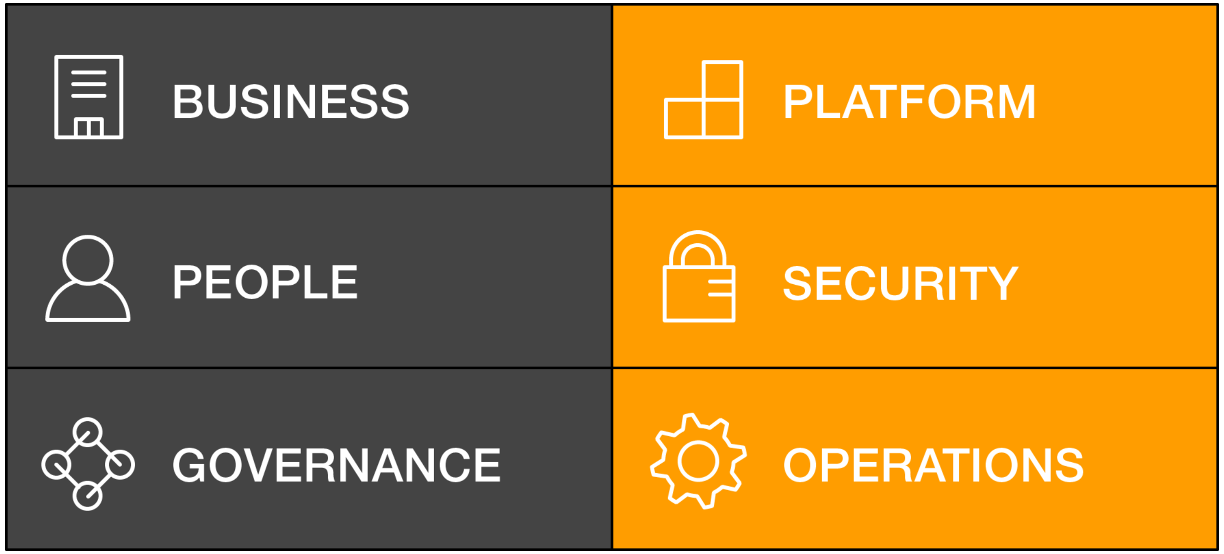
Section 3 key takeaways

- AWS is a secure cloud platform that offers a broad set of global cloud-based products called services that are designed to work together.
- There are many categories of AWS services, and each category has many services to choose from.
- Choose a service based on your business goals and technology requirements.
- There are three ways to interact with AWS services.

Section 4: Moving to the AWS Cloud – The AWS Cloud Adoption Framework (AWS CAF)

Module 1: Cloud Concepts Overview

AWS Cloud Adoption Framework (AWS CAF)



AWS CAF perspectives

- **AWS CAF provides guidance and best practices to help organizations** build a comprehensive approach to cloud computing across the organization and throughout the IT lifecycle to **accelerate successful cloud adoption**.
- AWS CAF is organized into **six *perspectives***.
- Perspectives consist of sets of ***capabilities***.

Six core perspectives

BUSINESS
PEOPLE
GOVERNANCE

Focus on **business** capabilities

PLATFORM
SECURITY
OPERATIONS

Focus on **technical** capabilities

Business perspective

BUSINESS

- IT finance
- IT strategy
- Benefits realization
- Business risk management

Business perspective capabilities

We must ensure that **IT is aligned with business needs**, and that IT investments can be traced to demonstrable business results.

Business managers, finance managers, budget owners, and strategy stakeholders

People perspective

PEOPLE

- Resource management
- Incentive management
- Career management
- Training management
- Organizational change management

People perspective capabilities

We must prioritize **training, staffing, and organizational changes** to build an agile organization.

Human resources, staffing, and people managers

Governance perspective

GOVERNANCE

- Portfolio management
- Program and project management
- Business performance measurement
- License management

Governance perspective capabilities

We must ensure that **skills and processes align IT strategy and goals with business strategy and goals** so the organization can maximize the business value of its IT investment and minimize business risks.

CIO, program managers, enterprise architects, business analysts, and portfolio managers

Platform perspective

PLATFORM

- Compute provisioning
- Network provisioning
- Storage provisioning
- Database provisioning
- Systems and solution architecture
- Application development

Platform perspective capabilities

We must **understand and communicate the nature of IT systems and their relationships**. We must be able to **describe the architecture of the target state environment** in detail.

CTO, IT managers, and solutions architects

Security perspective

SECURITY

- Identity and access management
- Detective control
- Infrastructure security
- Data protection
- Incident response

Security perspective capabilities

We must ensure that the organization **meets its security objectives**.

CISO, IT security managers, and IT security analysts

Operations perspective

OPERATIONS

- Service monitoring
- Application performance monitoring
- Resource inventory management
- Release management/ change management
- Reporting and analytics
- Business continuity/ Disaster recovery
- IT service catalog

Operations perspective capabilities

We align with and support the operations of the business, and **define how day-to-day, quarter-to-quarter, and year-to-year business will be conducted**.

IT operations managers and IT support managers

Section 4 key takeaways

- Cloud adoption is not instantaneous for most organizations and requires a thoughtful, deliberate strategy and alignment across the whole organization.
- The AWS CAF was created to help organizations develop efficient and effective plans for their cloud adoption journey.
- The AWS CAF organizes guidance into six areas of focus, called perspectives.
- Perspectives consist of sets of business or technology capabilities that are the responsibility of key stakeholders.

Module wrap-up

Module 1: Cloud Concepts Overview

Module summary

In summary, in this module you learned how to:

- Define different types of cloud computing models
- Describe six advantages of cloud computing
- Recognize the main AWS service categories and core services
- Review the AWS Cloud Adoption Framework

Sample exam question

Why is AWS more economical than traditional data centers for applications with varying compute workloads?

Choice	Response
A	Amazon Elastic Compute Cloud (Amazon EC2) costs are billed on a monthly basis.
B	Customers retain full administrative access to their Amazon EC2 instances.
C	Amazon EC2 instances can be launched on-demand when needed.
D	Customers can permanently run enough instances to handle peak workloads.

Additional resources

- What is AWS? YouTube video: https://www.youtube.com/watch?v=mZ5H8sn_2ZI&feature=youtu.be
- Cloud computing with AWS website: https://aws.amazon.com/what-is-aws/
- Overview of Amazon Web Services whitepaper: https://d1.awsstatic.com/whitepapers/aws-overview.pdf
- An Overview of the AWS Cloud Adoption Framework whitepaper: https://d1.awsstatic.com/whitepapers/aws_cloud_adoption_framework.pdf
- 6 Strategies for Migrating Applications to the Cloud AWS Cloud Enterprise Strategy blog post: https://aws.amazon.com/blogs/enterprise-strategy/6-strategies-for-migrating-applications-to-the-cloud/

Rest break

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Module 2: Cloud Economics and Billing

AWS Academy Cloud Foundations

Module overview

Topics

- Fundamentals of pricing
- Total Cost of Ownership
- AWS Organizations
- AWS Billing and Cost Management
- Technical Support

Activities

- AWS Pricing Calculator
- Support plans scavenger hunt

Demo

- Overview of the Billing Dashboard

Knowledge check

Module objectives

After completing this module, you should be able to:

- Explain the AWS pricing philosophy
- Recognize fundamental pricing characteristics
- Indicate the elements of total cost of ownership
- Discuss the results of the AWS Pricing Calculator
- Identify how to set up an organizational structure that simplifies billing and account visibility to review cost data.
- Identify the functionality in the AWS Billing Dashboard
- Describe how to use AWS Bills, AWS Cost Explorer, AWS Budgets, and AWS Cost and Usage Reports
- Identify the various AWS technical support plans and features

Section 1: Fundamentals of pricing

Module 2: Cloud Economics and Billing

AWS pricing model

Three fundamental drivers of cost with AWS

Compute

- Charged per hour/second*
- Varies by instance type

*Linux only

Storage

- Charged typically per GB

Data transfer

- Outbound is aggregated and charged
- Inbound has no charge (with some exceptions)
- Charged typically per GB

How do you pay for AWS?

Pay for what you use

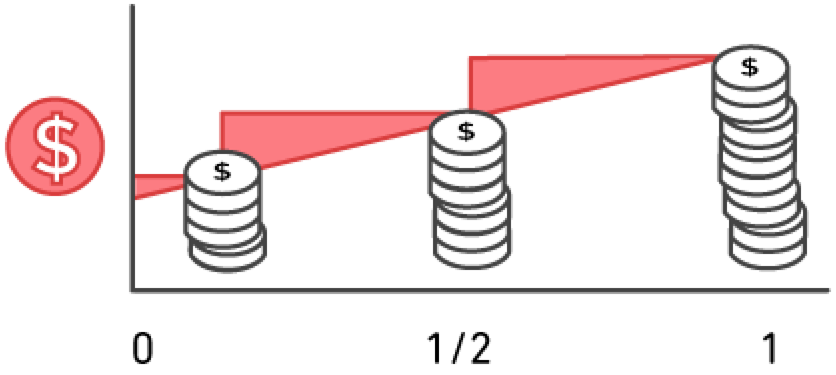
Pay less when you reserve

Pay less when you use more and as AWS grows

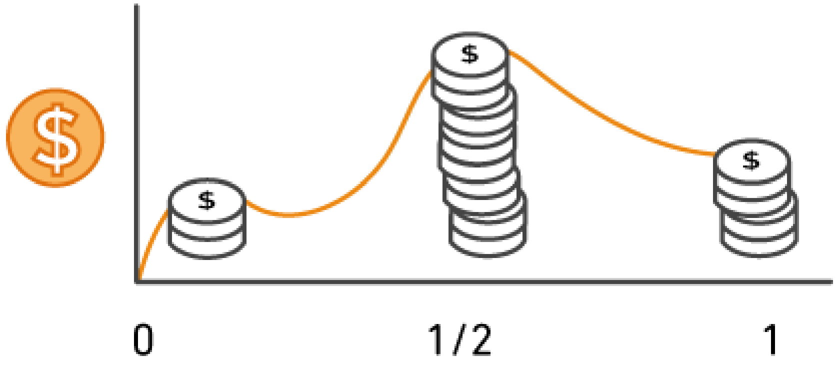
Pay for what you use

Pay only for the services that you consume, with no large upfront expenses.

On premises



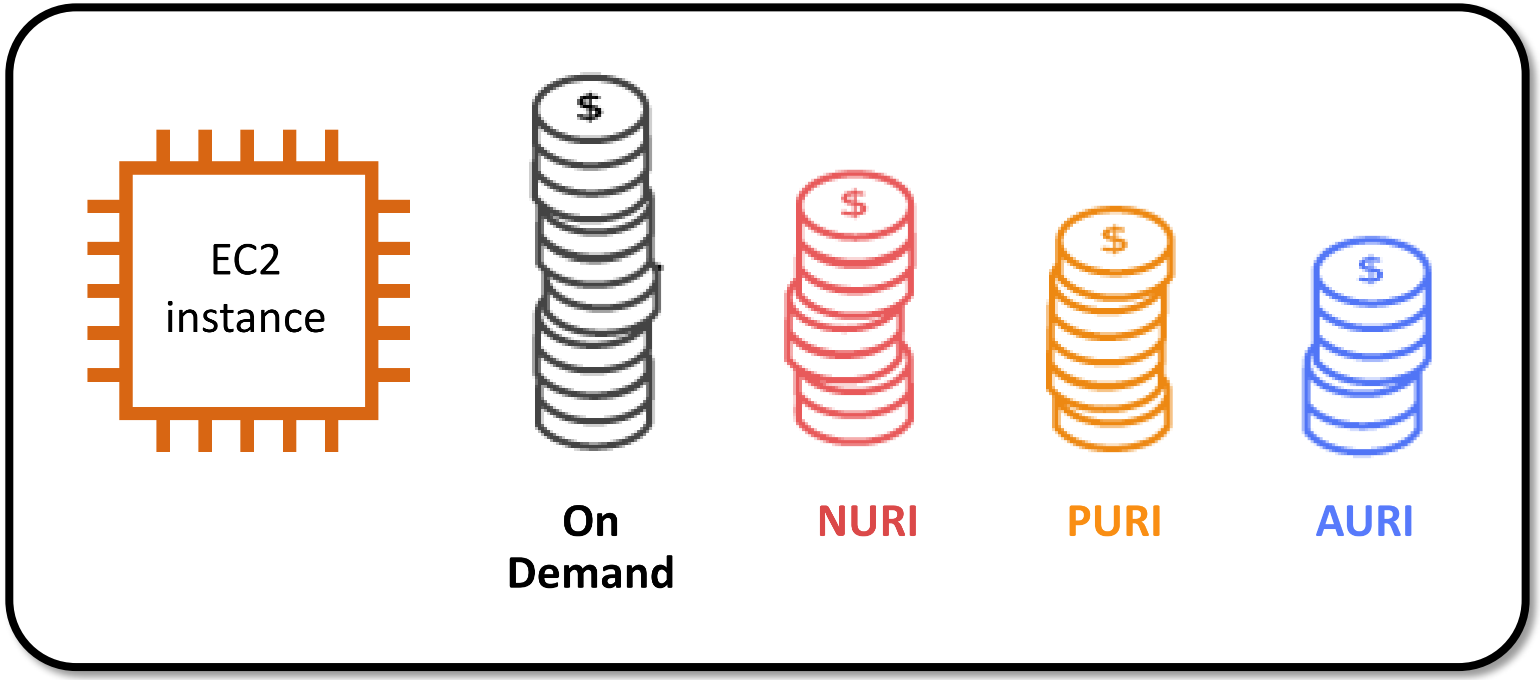
AWS



Pay less when you reserve

Invest in Reserved Instances (RIs):

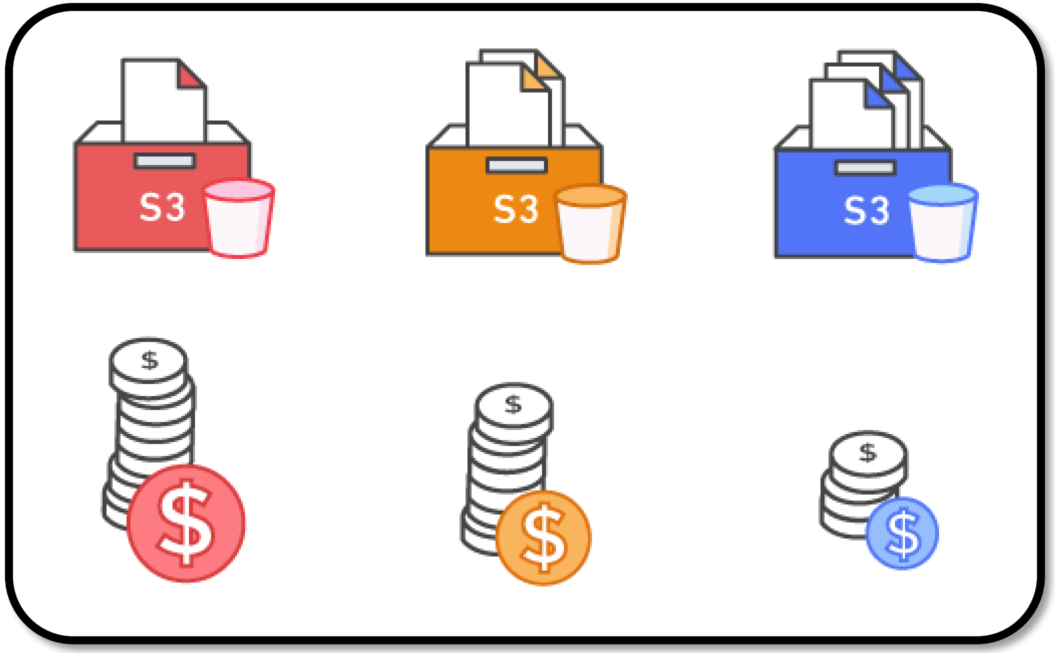
- Save up to 75 percent
- Options:
  - All Upfront Reserved Instance **(AURI)** → **largest discount**
  - Partial Upfront Reserved Instance **(PURI)** → **lower discounts**
  - No Upfront Payments Reserved Instance **(NURI)** → **smaller discount**



Pay less by using more

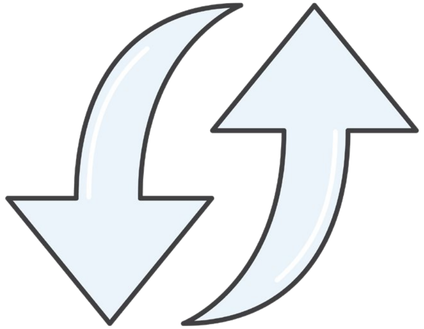
Realize volume-based discounts:

- **Savings** as usage increases.
- **Tiered pricing** for services like Amazon Simple Storage Service (Amazon S3), Amazon Elastic Block Store (Amazon EBS), or Amazon Elastic File System (Amazon EFS) → the more you use, the less you pay per GB.
- Multiple storage services deliver **lower** storage costs based on needs.



Pay even less as AWS grows

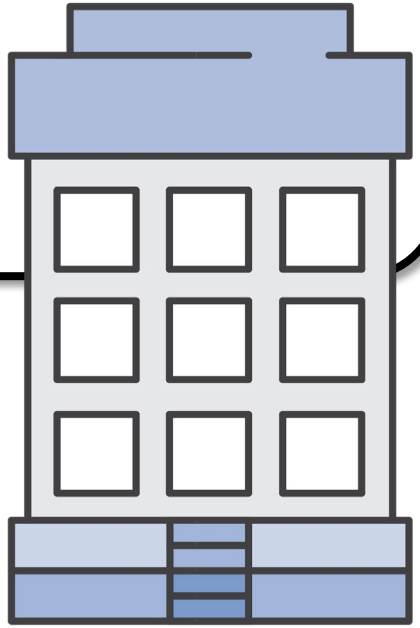
As AWS grows:



- AWS focuses on lowering cost of doing business.
- This practice results in AWS passing savings from economies of scale to you.
- Since 2006, AWS has **lowered pricing 75** times (as of September 2019).
- Future higher-performing resources replace current resources for no extra charge.

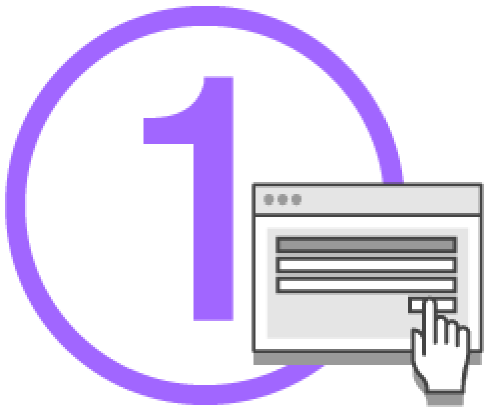
Custom pricing

- Meet varying needs through custom pricing.
- Available for high-volume projects with unique requirements.



AWS Free Tier

Enables you to gain free hands-on experience with the AWS platform, products, and services. Free for 1 year for new customers.



Sign up for an AWS account

Learn with 10-minute tutorials

Start building with AWS

Services with no charge

- Amazon VPC
- Elastic Beanstalk**
- Auto Scaling**
- AWS CloudFormation**
- AWS Identity and Access Management (IAM)

****Note**: There might be charges associated with other AWS services that are used with these services.

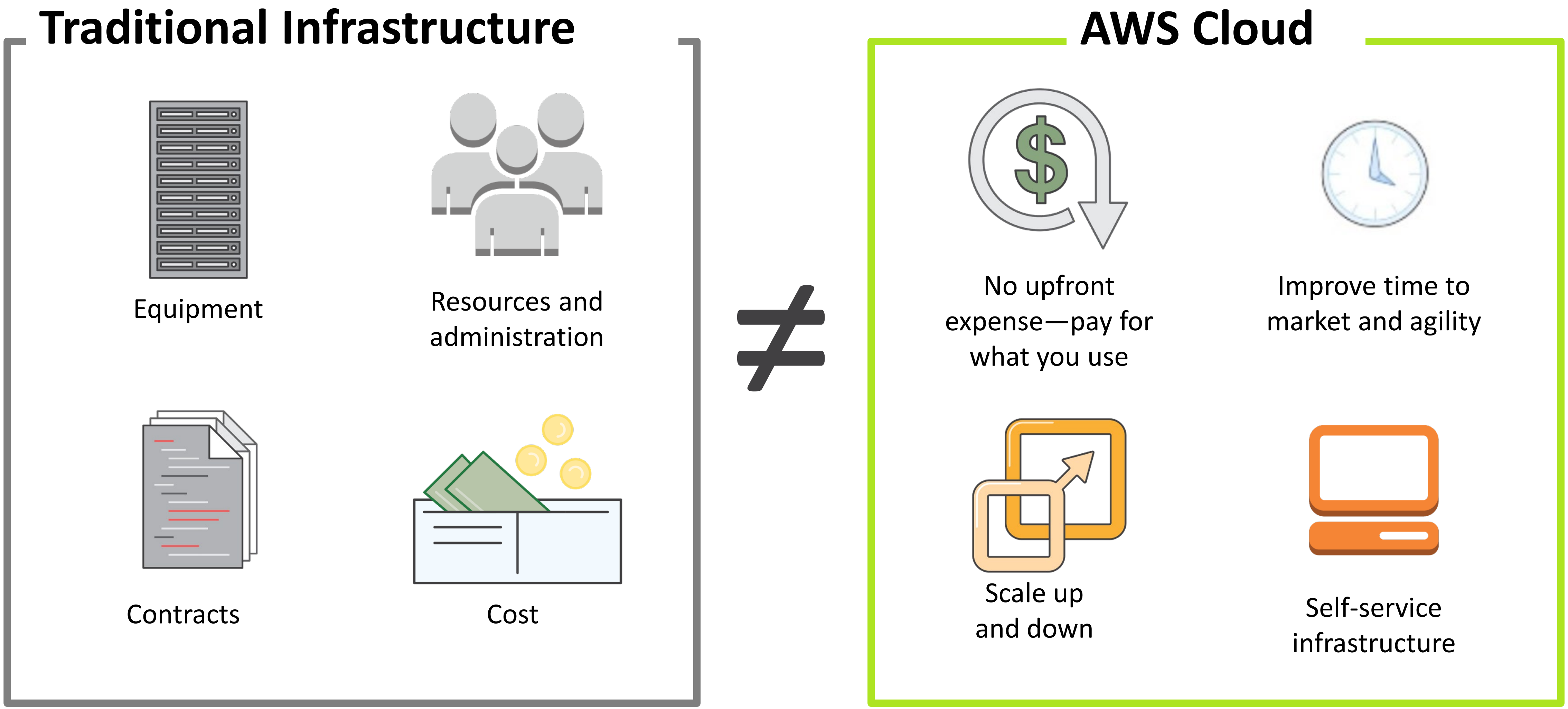
Key takeaways

- There is no charge (with some exceptions) for:
  - Inbound data transfer.
  - Data transfer between services within the same AWS Region.
- Pay for what you use.
- Start and stop anytime.
- No long-term contracts are required.
- Some services are free, but the other AWS services that they provision might not be free.

Section 2: Total Cost of Ownership

Module 2: Cloud Economics and Billing

On-premises versus cloud



What is Total cost of Ownership (TCO)?

Total Cost of Ownership (TCO) is the financial estimate to help identify direct and indirect costs of a system.

Why use TCO?

- To compare the costs of running an **entire infrastructure environment or specific workload** on-premises versus on AWS
- To budget and **build the business case** for moving to the cloud

TCO considerations

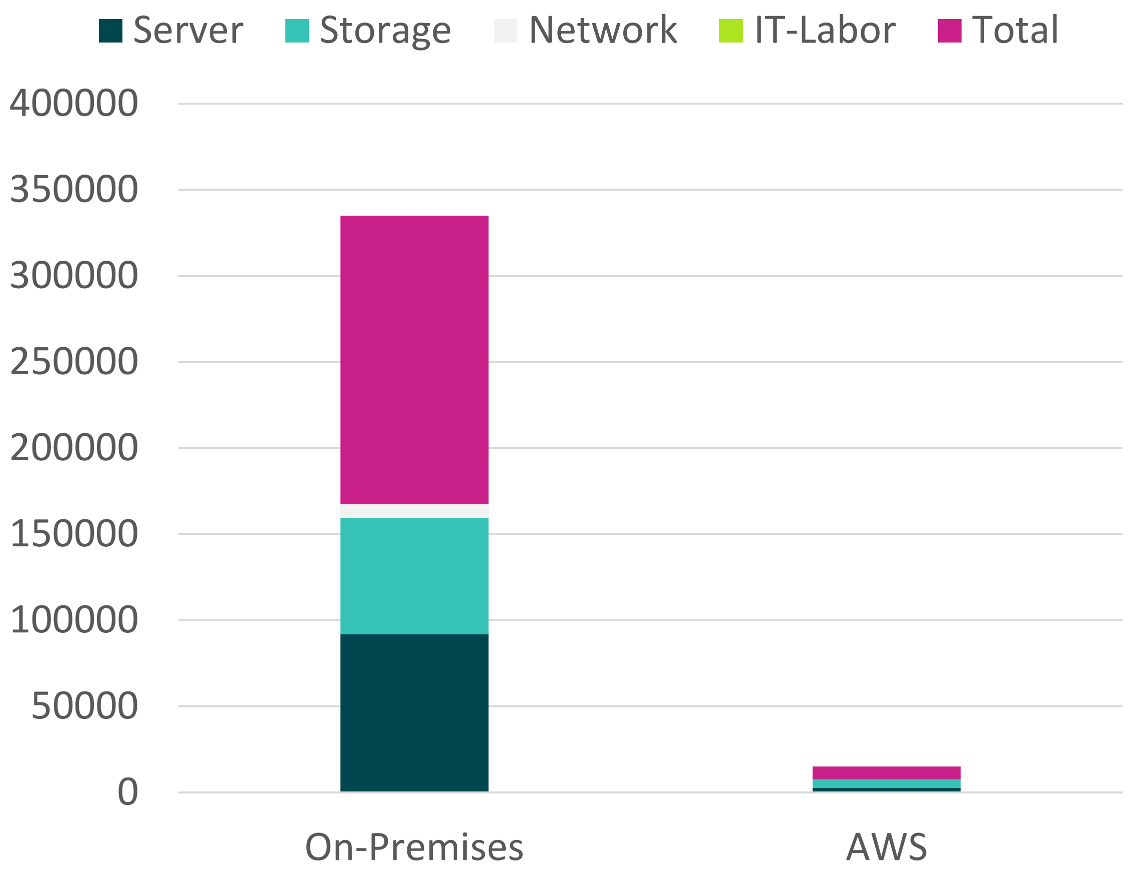
1 Server Costs	Hardware: Server, rack chassis power distribution units (PDUs), top-of-rack (TOR) switches (and maintenance)	Software: Operating system (OS), virtualization licenses (and maintenance)	Facilities cost		
			Space	Power	Cooling
2 Storage Costs	Hardware: Storage disks, storage area network (SAN) or Fibre Channel (FC) switches	Storage administration costs	Facilities cost		
			Space	Power	Cooling
3 Network Costs	Network hardware: Local area network (LAN) switches, load balancer bandwidth costs	Network administration costs	Facilities cost		
			Space	Power	Cooling
4 IT Labor Costs	Server administration costs				

On-premises versus all-in-cloud

You could save up to **96 percent** a year by moving your infrastructure to AWS.
Your 3-year total savings would be **$159,913**.

3-Year Total Cost of Ownership		
	On-Premises	AWS
Server	$91,922	$2,547
Storage	$67,840	$4,963
Network	$7,660	$-------
IT – Labor	$ -------- --	$-------
Total	$167, 422	$7,509

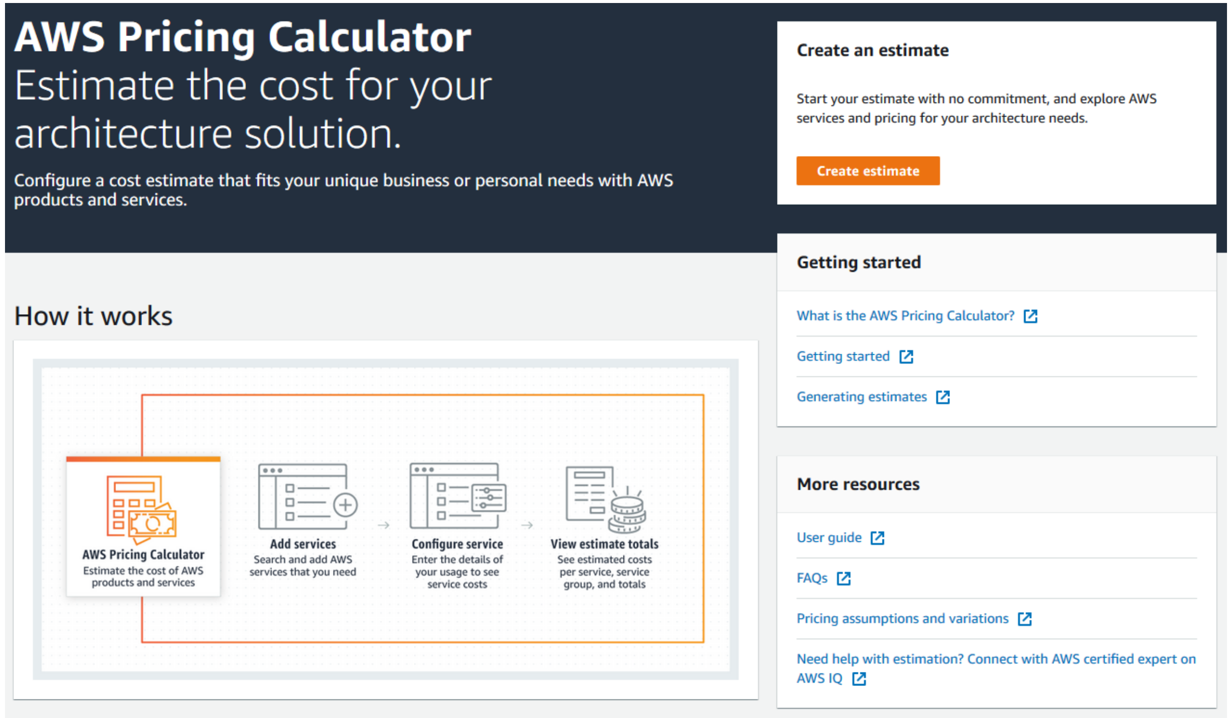
AWS cost includes business-level support and a 3-year PURI EC2 instance



AWS Pricing Calculator

Use the AWS Pricing Calculator to:

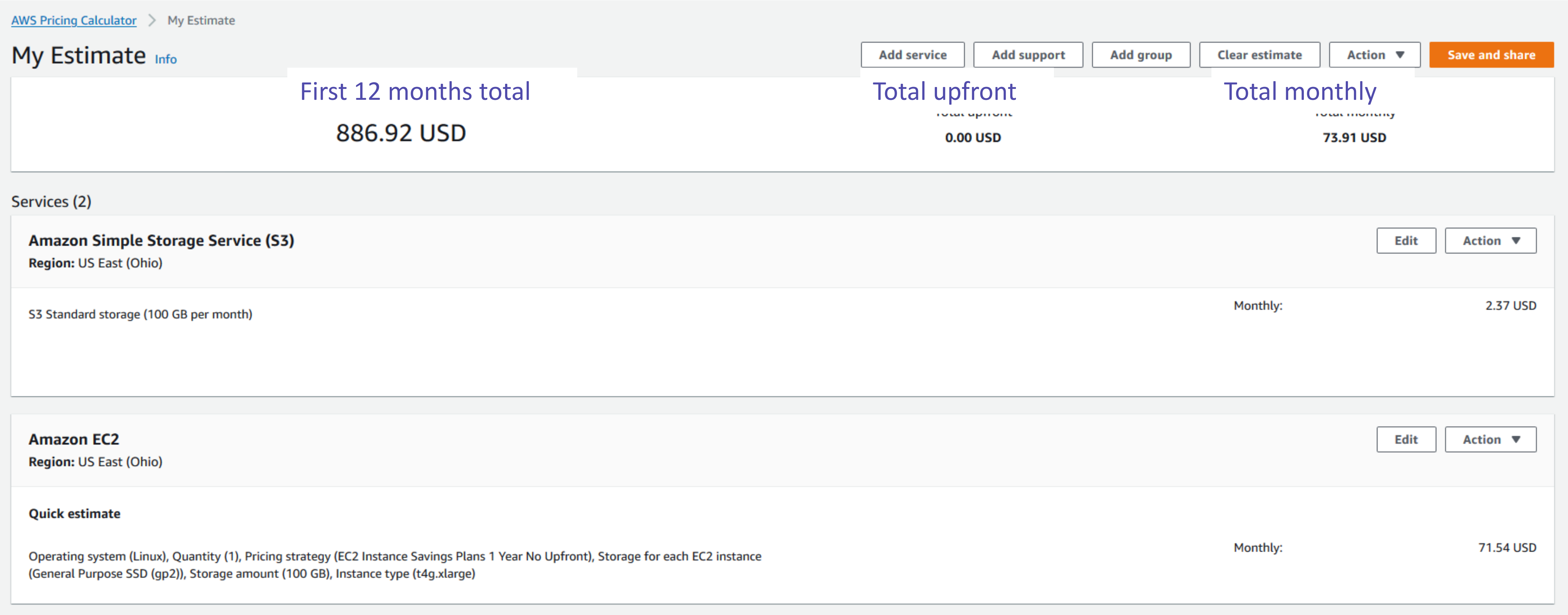
- Estimate monthly costs
- Identify opportunities to reduce monthly costs
- Model your solutions before building them
- Explore price points and calculations behind your estimate
- Find the available instance types and contract terms that meet your needs
- Name your estimate and create and name groups of services



Access the AWS Pricing Calculator

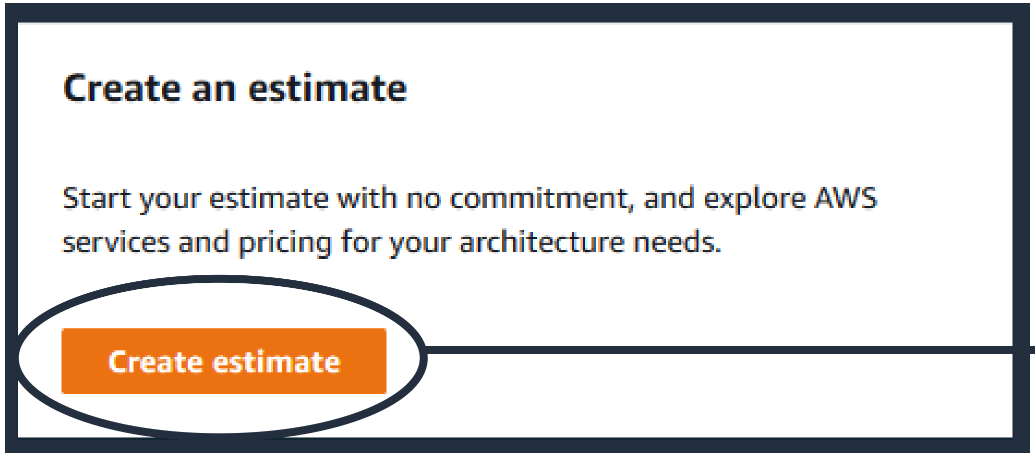
Reading an estimate

Your estimate is broken into: first 12 months total, total upfront, and total monthly.

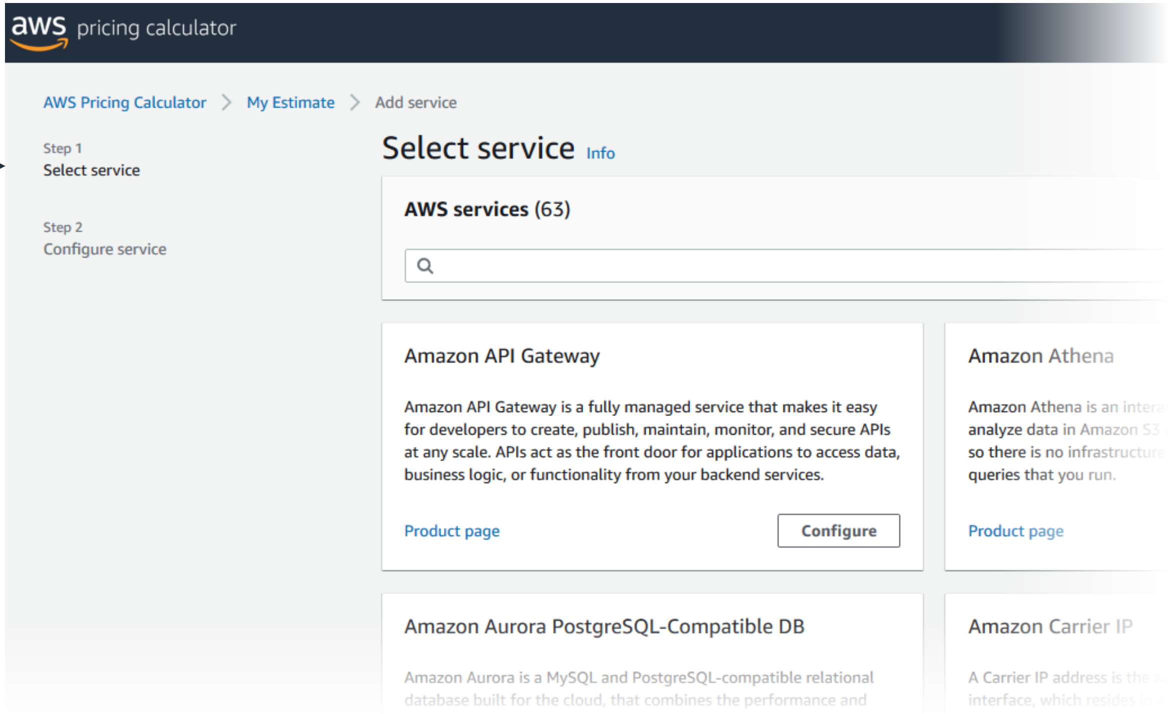


Activity: AWS Pricing Calculator activity

- Break up into groups of four or five and use the AWS Pricing Calculator and specifications provided to develop a cost estimate.
- Be prepared to report your findings back to the class.



AWS Pricing calculator website



Additional benefit considerations

Hard benefits

- Reduced spending on compute, storage, networking, security
- Reductions in hardware and software purchases (capex)
- Reductions in operational costs, backup, and disaster recovery
- Reduction in operations personnel

Soft Benefits

- Reuse of service and applications that enable you to define (and redefine solutions) by using the same cloud service
- Increased developer productivity
- Improved customer satisfaction
- Agile business processes that can quickly respond to new and emerging opportunities
- Increase in global reach

Case study: Total Cost Of Ownership (1 of 6)

Background:

- Growing global company with over 200 locations
- 500 million customers, $3 billion annual revenue

Case study: Total Cost of Ownership (2 of 6)

Delaware North

Background:

- Growing global company with over 200 locations
- 500 million customers, $3 billion annual revenue

Challenge:

- Meet demand to rapidly deploy new solutions
- Constantly upgrade aging equipment

Case study: Total Cost of Ownership (3 of 6)

Delaware North

Background:

- Growing global company with over 200 locations
- 500 million customers, $3 billion annual revenue

Challenge:

- Meet demand to rapidly deploy new solutions
- Constantly upgrade aging equipment

Criteria:

- Broad solution to handle all workloads
- Ability to modify processes to improve efficiency and lower costs
- Eliminate busy work (such as patching software)
- Achieve a positive return on investment (ROI)

Case study: Total Cost of Ownership (4 of 6)

Background:

- Is a growing global company with over 200 locations
- Have 500 million customers, $3 billion (USD) annual revenue

Challenge:

- Meet demand to rapidly deploy new solutions
- Constantly upgrade aging equipment

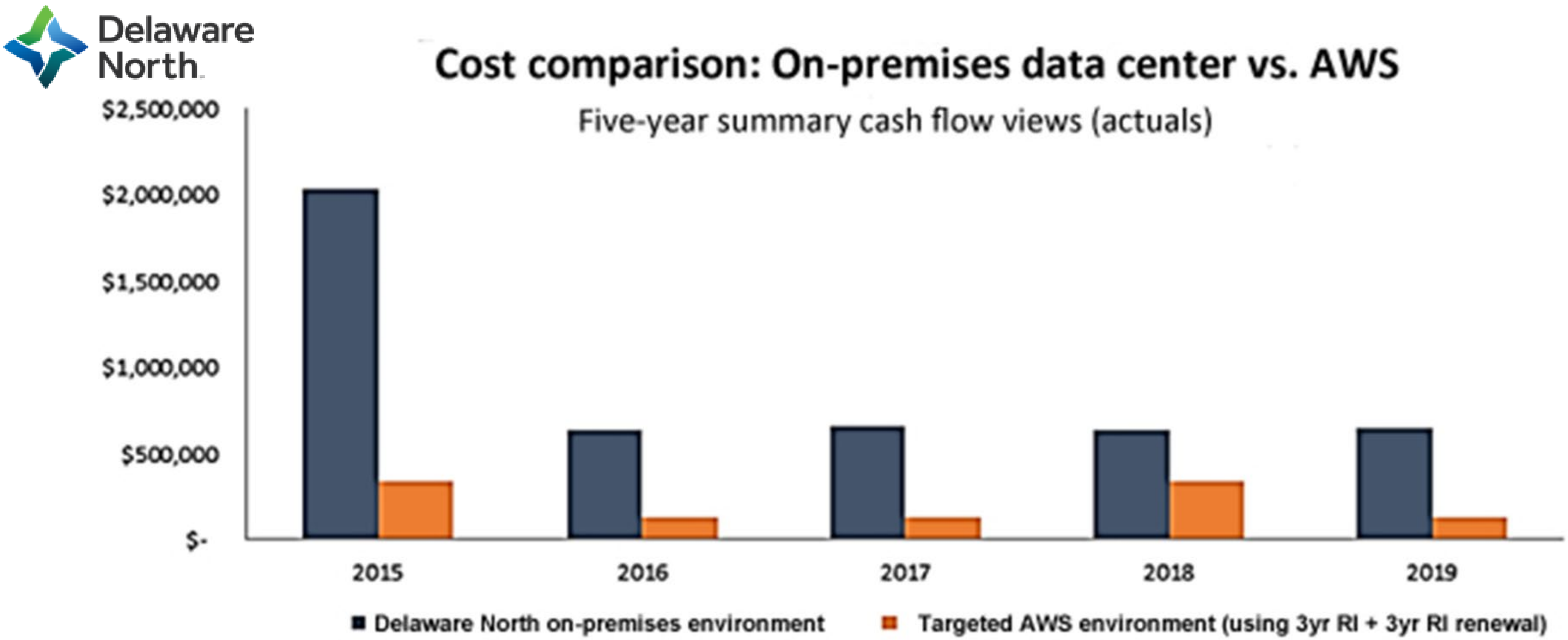
Criteria:

- Have a broad solution to handle all workloads
- Be able to modify processes to improve efficiency and lower costs
- Eliminate busy work (such as patching software)
- Achieve a positive return on investment (ROI)

Solution:

- Moved their on-premises data center to AWS
  - Eliminated 205 servers (90 percent)
  - Moved nearly all applications to AWS
- Used 3-year Amazon EC2 Reserved Instances

Case study: Total Cost of Ownership (5 of 6)



Case study: Total Cost of Ownership (6 of 6)

Delaware North

Results:

Business Goals:
Growth
Enhanced 24/7 business
Operational efficiency

Resource optimization

- Robust security compliance
- Enhanced disaster recovery
- Increased computing capacity

Speed to market

- One day to provision new businesses
- Just minutes to push out a service

Operational efficiency

- Continuous cost optimization and reduction

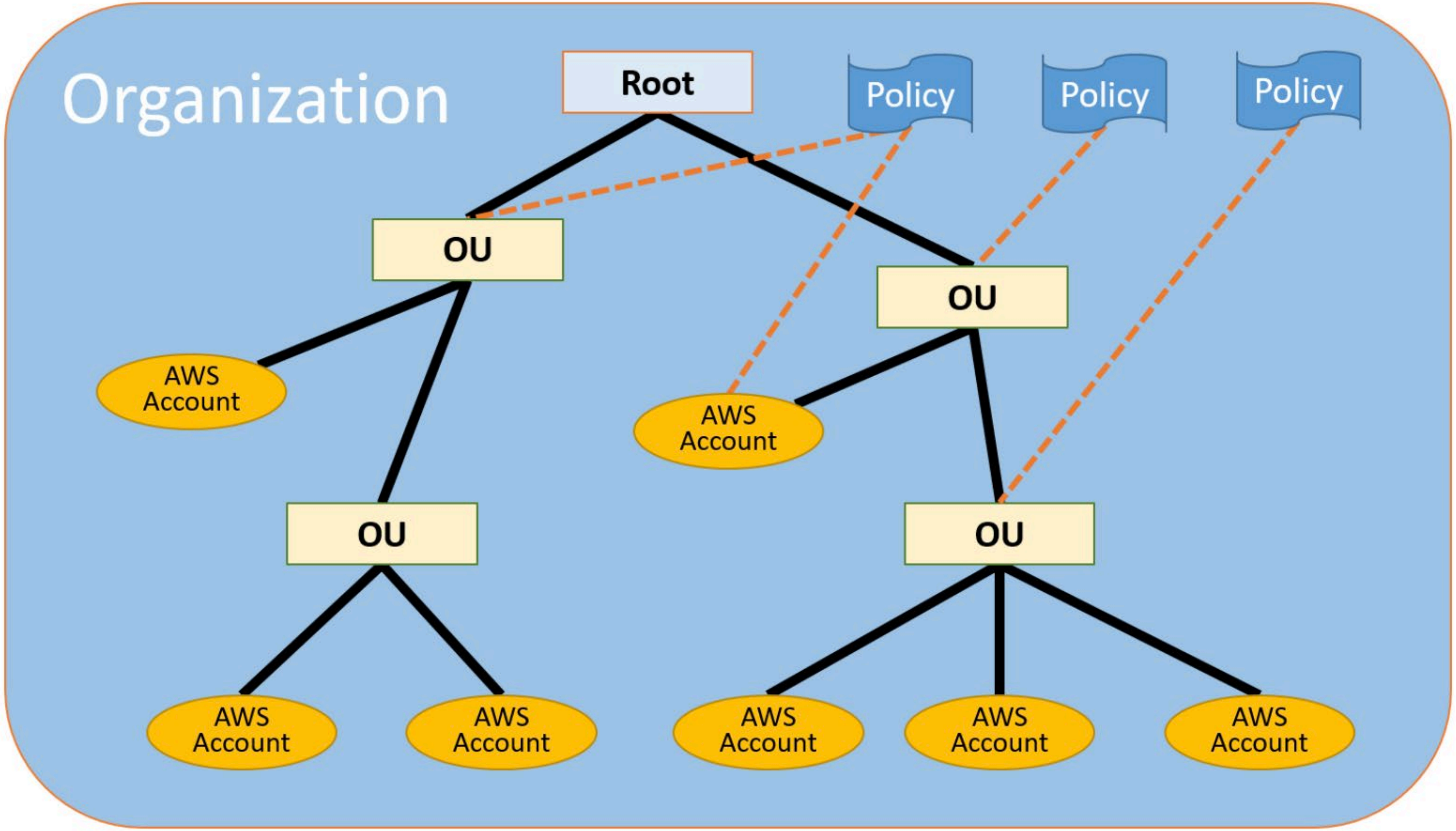
Section 3: AWS Organizations

Module 2: Cloud Economics and Billing

Introduction to AWS Organizations

AWS Organizations

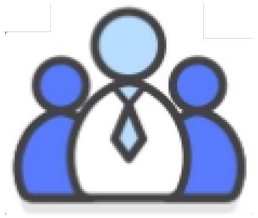
AWS Organizations terminology



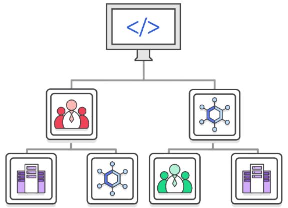
*Organizational Units (OUs)

Key features and benefits

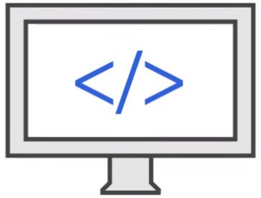
AWS Organizations



Policy-based account management



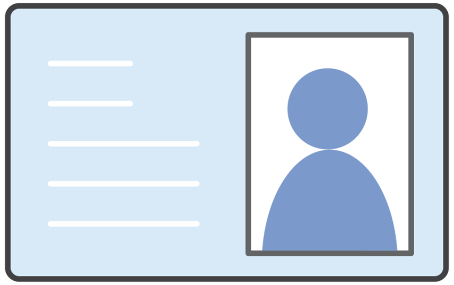
Group based account management



Application programming interfaces (APIs) that automate account management

Consolidated billing

Security with AWS Organizations

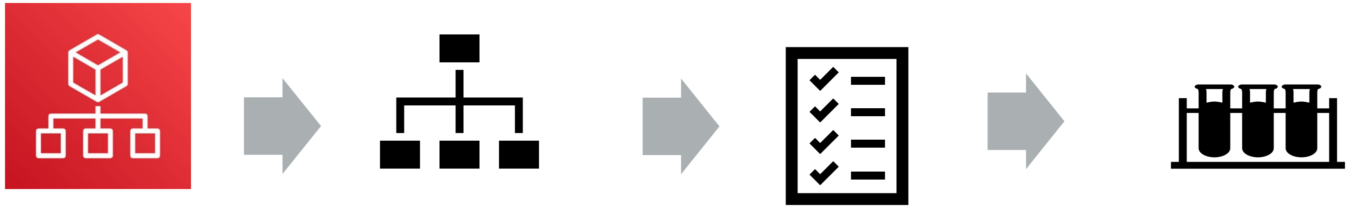


Control access with AWS Identity and Access Management (IAM).

IAM policies enable you to allow or deny access to AWS services for users, groups, and roles.

Service control policies (SCPs) enable you to allow or deny access to AWS services for individuals or group accounts in an organizational unit (OU).

Organizations setup



Step 1

Create Organization

Step 2

Create organizational units

Step 3

Create service control policies

Step 4

Test restrictions

Limits of AWS Organizations

Limits		
Limits on Names	Names must be composed of Unicode characters.	
	Names must not exceed 250 characters in length.	
Maximum and Minimum Values	Number of AWS accounts	Varies. Note: An invitation sent to an account counts against this limit.
	Number of roots	1
	Number of OUs	1,000
	Number of policies	1,000
	Maximum size of a service control policy document	5,120 bytes
	Maximum nesting of OUs in a root	5 levels of OUs under a root
	Invitations sent per day	20
	Number of member accounts you can create concurrently	Only five can be in progress at one time
	Number of entities to which you can attach a policy	Unlimited

Accessing AWS Organizations

AWS Organizations

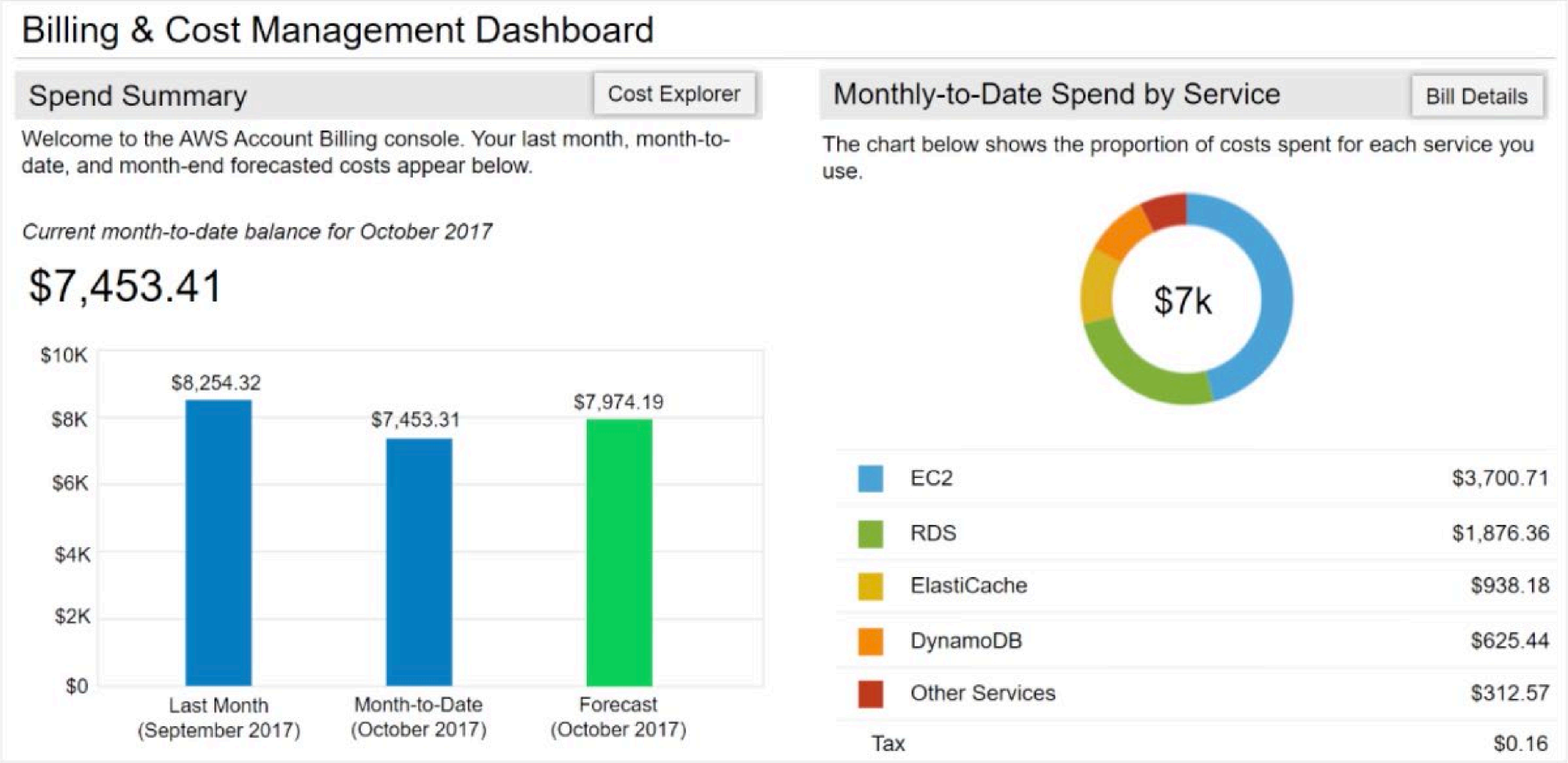
- AWS Management Console
- AWS Command Line Interface (AWS CLI) tools
- Software development kits (SDKs)
- HTTPS Query application programming interfaces (API)

Section 4: AWS Billing and Cost Management

Module 2: Cloud Economics and Billing

Introducing AWS Billing and Cost Management

AWS Billing Dashboard



Tools

AWS Budgets

AWS Cost and Usage Report

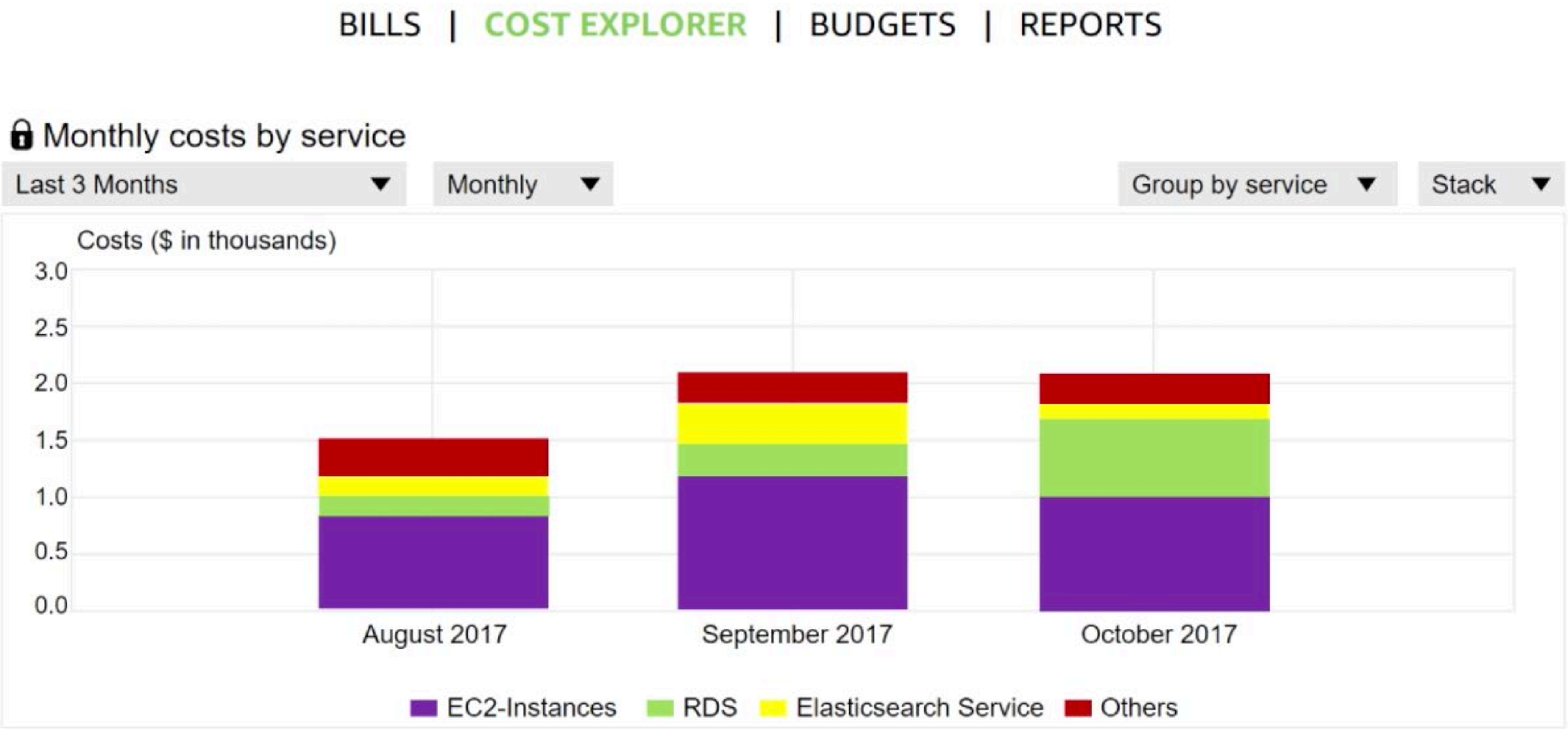
AWS Cost Explorer

Monthly bills

BILLS | COST EXPLORER | BUDGETS | REPORTS

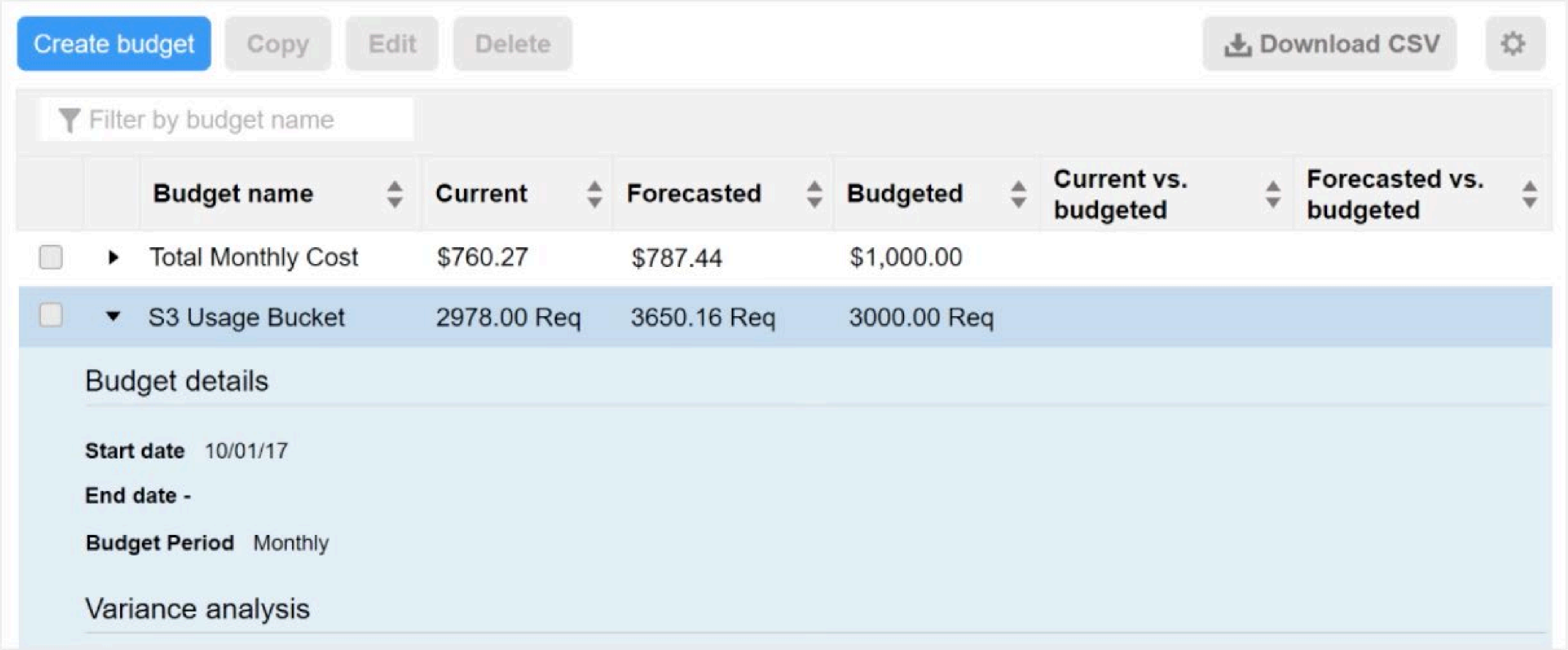
Total		$7,453.41 USD
AWS Marketplace Charges		**$15.00**
▼ Usage Charges and Recurring Fees		$15.00
Invoice 32342548 – AWS Service Charges: Usage charge for this statement period	2017-10-10	$15.00
AWS Service Charges		**$7,438.41**
▼ Usage Charges and Recurring Fees		$7,414.41
Invoice 32342513 – AWS Service Charges: Usage charge for this statement period	2017-10-10	$7,414.41
▼ Usage Charges and Recurring Fees		$24.00
Invoice 32342507 – AWS Service Charges: Subscription charge	2017-10-10	$24.00

Cost Explorer



Forecast and track costs

BILLS | COST EXPLORER | BUDGETS | REPORTS



Cost and usage reporting

BILLS | COST EXPLORER | BUDGETS | REPORTS

Product Code	Usage Type	Operation	Availability Zone	Usage Amount	Currency Code	Line Item Description
Amazon S3	Requests – Tier 1	ListAllMyBuckets		2	USD	$0.00 per request – PUT, COPY, POST, LIST under the global free tier
Amazon EC2	USW2-Boxusage:t2.micro	Runinstnaces:0002	us-west-2a	1	USD	$0.00 per Windows t2.micro instance-hour under monthly free tier
Amazon S3	Requests – Tier 1	ListAllMyBuckets		2	USD	$0.00 per request – PUT, COPY, POST, LIST under the global free tier
Amazon EC2	USW2-Boxusage:t2.micro	Runinstnaces:0002	us-west-2a	1	USD	$0.00 per Windows t2.micro instance-hour under monthly free tier
Amazon S3	Requests – Tier 1	ListAllMyBuckets		2	USD	$0.00 per request – PUT, COPY, POST, LIST under the global free tier
Amazon S3	Requests – Tier 1	ListAllMyBuckets		2	USD	$0.00 per request – PUT, COPY, POST, LIST under the global free tier

Recorded demo: Amazon Billing dashboard

Billing dashboard demonstration

Getting Started with AWS Billing & Cost Management

- Manage your costs and usage using AWS Budgets
- Visualize your cost drivers and usage trends via Cost Explorer
- Dive deeper into your costs using the Cost and Usage Reports with Athena integration
- **Learn more:** Check out the AWS What's New webpage

Do you have Reserved Instances (RIs)?

- Access the RI Utilization & Coverage reports—and RI purchase recommendations—via Cost Explorer.

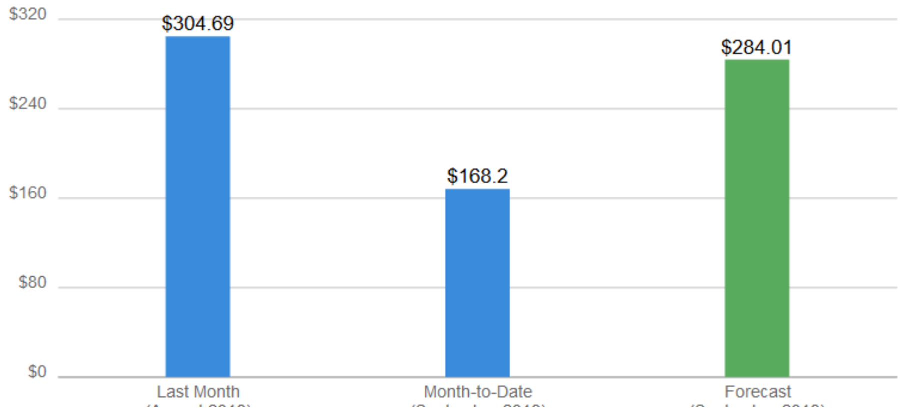
Spend Summary

Cost Explorer

Welcome to the AWS Billing & Cost Management console. Your last month, month-to-date, and month-end forecasted costs appear below.

Current month-to-date balance for September 2019

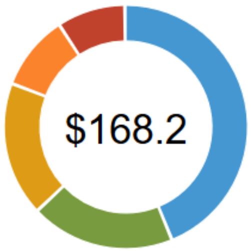
$168.20



Month-to-Date Spend by Service

Bill Details

The chart below shows the proportion of costs spent for each service you use.



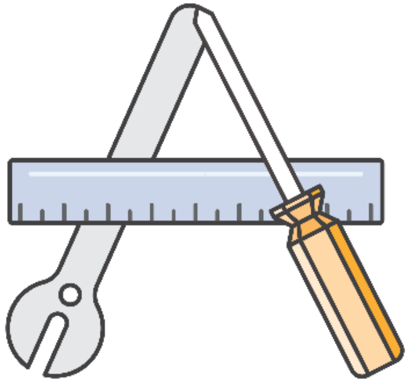
ES	$74.52
DatabaseMigrationSvc	$32.12
SageMaker	$29.99
EC2	$16.59
Other Services	$14.98
Tax	$0.00
Total	**$168.20**

Section 5: Technical support

Module 2: Cloud Economics and Billing

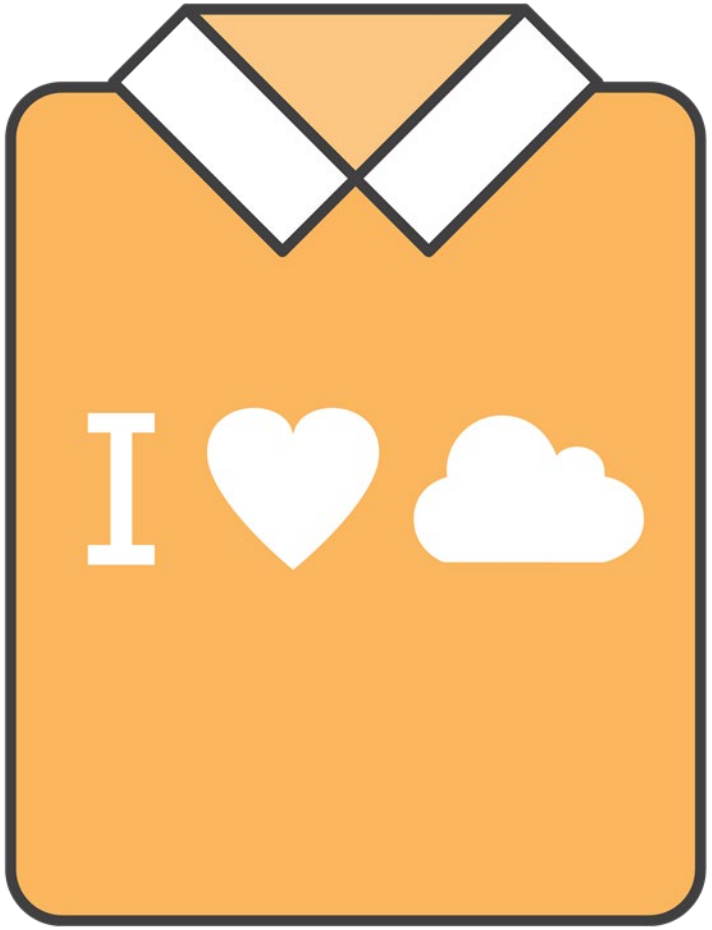
AWS support (1 of 2)

- Provide unique combination of tools and expertise:
  - AWS Support
  - AWS Support Plans
- Support is provided for:
  - Experimenting with AWS
  - Production use of AWS
  - Business-critical use of AWS



AWS support (2 of 2)

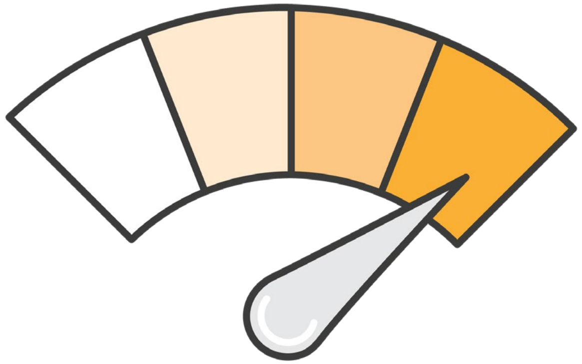
- Proactive guidance :
  - Technical Account Manager (TAM)
- Best practices :
  - AWS Trusted Advisor
- Account assistance :
  - AWS Support Concierge



Support plans

AWS Support offers four support plans:

- **Basic Support** – Resource Center access, Service Health Dashboard, product FAQs, discussion forums, and support for health checks
- **Developer Support**: Support for early development on AWS
- **Business Support**: Customers that run production workloads
- **Enterprise Support**: Customers that run business and mission-critical workloads



Case severity and response times

	Critical	Urgent	High	Normal	Low
Basic	No Case Support				
Developer Plan (Business hours)				12 hours or less	24 hours or less
Business Plan (24/7)		1 hour or less	4 hours or less	12 hours or less	24 hours or less
Enterprise Plan (24/7)	15 minutes or less	1 hour or less	4 hours or less	12 hours or less	24 hours or less

Activity: Support plan scavenger hunt

- Break up into groups of four or five and develop a recommendation for the best support plan for one of the business cases that are provided.

- Be prepared to report your findings back to the class.

Module wrap-up

Module 2: Cloud Economics and Billing

Module summary

- Explored the fundamental of AWS pricing
- Reviewed TCO concepts
- Reviewed an AWS Pricing Calculator estimate
- Reviewed the Billing dashboard
- Reviewed Technical Support options and costs

Sample exam question

Which AWS service provides infrastructure security optimization recommendations?

Choice	Response
A	AWS Price List Application Programming Interface (API)
B	Reserved Instances
C	AWS Trusted Advisor
D	Amazon Elastic Compute Cloud (Amazon EC2) Spot Fleet

Additional resources

- AWS Economics Center: http://aws.amazon.com/economics/
- AWS Pricing Calculator: https://calculator.aws/#/
- Case studies and research: http://aws.amazon.com/economics/
- Additional pricing exercises: https://dx1572sre29wk.cloudfront.net/cost/

Thank you

Corrections, feedback, or other questions?
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